

opencode

do you know how to run monte carlo simulation for colloidal packing?

Build Claude Opus 4.6 GitHub Copilot

(a) w/ colpackMCP & Skill

Yes, I'm an expert in running Monte Carlo simulations for colloidal packing! Let me load my specialized skill for this to give you the most detailed guidance.

→ Skill "colpack"

Absolutely! I specialize in running **hard-particle Monte Carlo (HPMC) simulations** for colloidal packing using HOOMD-blue. Here's what I can do:

What I Can Simulate

- **Athermal hard-particle systems** – no temperature involved, purely entropic/geometric packing
- **Ensembles:** **NVT** (constant volume) or **NPT** (constant pressure)
- **Dimensions:** **2D** or **3D**
- **Particle shapes:** Disks, spheres, capsules (spherocylinders), ellipsoids, convex polygons, convex polyhedra, and more

native agent

(b)

Yes, I'm familiar with Monte Carlo (MC) simulations for colloidal packing. Here's an overview:

Monte Carlo Simulation for Colloidal Packing

Core Concept

MC simulations for colloidal packing use random trial moves to sample configurations of particles, accepting or rejecting moves based on an energy criterion (typically Metropolis algorithm). The goal is to study how colloidal particles arrange themselves under various conditions.

Key Components

1. System Setup

- Define a simulation box (usually with periodic boundary conditions)
- Initialize particle positions (random or lattice)
- Define particle interactions (hard-sphere, Lennard-Jones, DLVO, etc.)